Hi1910B / Atlas 200I A2 — Analysis and Documentation Links

Compiled from Huawei's official Atlas 200I A2 and Atlas 200I DK A2 documentation · October 2026

Direct download URLs

Devkit-level documentation

- AscendCL Application Development Guide (Python): https://www.hiascend.com/doc_center/source/zh/Atlas200IDKA2DeveloperKit/23.0.RC2/Application%20Development%20Guide/aadgp/AscendCL%20%E5%BA%94%E7%94%A8%E5%BC%80%E5%8F%91%E6%8C%87%E5%8D%97%EF%BC%88Python%EF%BC%89.pdf
- AscendC Operator Development Guide: https://www.hiascend.com/doc_center/source/zh/Atlas200IDKA2DeveloperKit/23.0.RC2/operatordev/Ascendcopdevg/Atlas%20200I%20DK%20A2%20%E5%BC%80%E5%8F%91%E8%80%85%E5%A5%97%E4%BB%B6%2023.0.RC3%20AscendC%E7%AE%97%E5%AD%90%E5%BC%80%E5%8F%91%E6%8C%87%E5%8D%97%2001.pdf

Module-level documentation

- Atlas 200I A2 Accelerator Module User Guide (specs, mechanical, installation): https://www.hiascend.com/doc_center/source/zh/Atlas%20200I%20A2/24.1.RC1/intro/userguide/Atlas%20200I%20A2%20%E5%8A%A0%E9%80%9F%E6%A8%A1%E5%9D%97%20%E7%94%A8%E6%88%B7%E6%8C%87%E5%8D%97.pdf
- Atlas 200I A2 Accelerator Module Driver Development Guide (names the hi1910B device tree files directly): https://www.hiascend.com/doc_center/source/zh/Atlas%20200I%20A2/23.0.0/RC/driverdevelopmentguide/Atlas%20200I%20A2%20%E5%8A%A0%E9%80%9F%E6%A8%A1%E5%9D%97%2023.0.x(23.0.3%E5%8F%8A%E5%85%B6%E7%B3%BB%E5%88%97%E7%89%88%E6%9C%AC)%20%E9%A9%B1%E5%8A%A8%E5%BC%80%E5%8F%91%E6%8C%87%E5%8D%97%2007.pdf

Note: no standalone chip-level datasheet for the Hi1910B SoC itself was found anywhere, official or third-party. Huawei documents the Atlas module and devkit as products; the SoC inside is not documented or sold as a separate component.

Key finding: which hardware variant is actually in the devkit

The Atlas 200I A2 module ships in four hardware configurations, distinguished by Board ID:

Board ID	Config	AI core clock	CPU clock	FP16	INT8	Memory speed
0x42	20 TOPS, 12GB	1.224 GHz	1.6 GHz	10 TFLOPS	20 TOPS	4266 Mbps
0x43	20 TOPS, 8GB	1.224 GHz	1.6 GHz	10 TFLOPS	20 TOPS	4266 Mbps
0x48	20 TOPS, 4GB	1.224 GHz	1.6 GHz	10 TFLOPS	20 TOPS	4266 Mbps
0x44	8 TOPS, 4GB	0.5 GHz	1.0 GHz	4 TFLOPS	8 TOPS	3200 Mbps

All four variants use the same silicon design — one DaVinci V300 AI core and four TaiShan V200M CPU cores — the difference is clock speed, not core count.

The driver development guide's device-tree naming labels the 8-TOPS/4GB variant as "M100," and the file `hi1910B-asic-M100-B51.dts` is the one used for the Atlas 200I DK A2 developer kit. Since the devkit's published spec sheet lists exactly 4 cores, 8 TOPS INT8, and 4GB memory, this confirms the Hi1910B in the devkit is the **0x44 configuration** — the lowest-clocked bin of this module family, running at roughly 40% of the AI-core clock and 60% of the CPU clock of the flagship 20 TOPS parts. Most third-party benchmarks and online specs for "Atlas 200I A2" describe the 20 TOPS variant, so performance comparisons should account for this gap (roughly 2.4x on raw INT8 throughput).

Full specifications (8 TOPS / 4GB variant, Board ID 0x44)

Compute

- AI core: 1x DaVinci V300 @ 0.5 GHz
- CPU: 4x TaiShan V200M @ 1.0 GHz
- Throughput: 4 TFLOPS FP16, 8 TOPS INT8
- Memory: LPDDR4X, 3200 Mbps, ECC supported, 4GB, 64-bit bus

Storage

- 64MB built-in SPI flash
- External eMMC interface, up to eMMC 5.1 / HS400 mode
- Up to 4 external SATA 3.0 ports, backward compatible to SATA 2.0/1.0, with Port Multiplier support

Video codec

- H.264/H.265 decode: 20 channels at 1080p30, or 2 channels at 4K@75fps (YUV420)
- H.264/H.265 encode: 12 channels at 1080p30, or 2 channels at 4K@50fps (YUV420)
- JPEG: decode 512fps at 1080p (256fps max per single channel), encode 256fps at 1080p
- Supported resolution range: 32x32 up to 16384x16384

High-speed interfaces

- 8 shared SerDes lanes, configurable across: SGMII/1000BASE-R (1.25 Gbps), USB 3.0 (5 Gbps), SATA 3.0 (6 Gbps, backward compatible to 2.0/1.0), PCIe Gen3 (8 Gbps, backward compatible to Gen2/Gen1)
- PCI IDs: Vendor 0x19E5, Device 0xD105, Subsystem Vendor 0x0200, Subsystem Device 0x0100

Low-speed peripherals

The document has an internal inconsistency worth flagging: the summary spec table lists UART:5, I2C:4, SPI:2, CAN:4, while the detailed interface section elsewhere in the same guide describes 7 UART channels, 7 I2C channels (1 slave + 6 master), and 6 SPI controllers. This is unresolved in Huawei's own documentation — treat the detailed section as describing full silicon capability, and check the Atlas 200I DK A2 pinout/PinMux documentation for what is actually exposed on the physical 40-pin connector, since not every SoC-level interface reaches a connector pin.

Camera and display I/O

- 2x MIPI CSI-2 interfaces: up to 4 simultaneous sensors, max 4096x2160@45fps, RAW8/10/12/14 and YUV420/422 formats, up to 4 virtual channels, 2-frame WDR support
- MIPI-DSI output: up to 1920x1080@60fps, RGB 16/18/24-bit and YUV420/422 formats
- 2x HDMI 2.0 outputs, up to 3840x2160@60fps
- 2x I2S audio interfaces, 1x analog audio line-out, 1x analog audio input (supports differential or single-ended)

Other interfaces

- 2x RGMII network ports
- 1x SDIO (SDXC/SDHC, 3.3V/1.8V)
- 2x PWM

Power and mechanical

- Input voltage: 12V
- Typical power draw: 21W (this 8 TOPS/4GB variant) vs. 23.2–25W for the 20 TOPS variants
- Dimensions: 82mm x 60mm x 7mm
- Weight: 80g
- Connector: 314-pin MXM gold-finger connector, plus two 51-pin FPC connectors

Environmental

- Operating junction temperature: −20°C to +103°C
- Storage temperature: −40°C to +85°C
- Operating humidity: 5–90% (non-condensing)
- Altitude: usable under 5000m, with 1°C temperature derating per 220m above 1800m

Software

- Supported OS: openEuler 22.03, Ubuntu 22.04

Bottom line

The part being analyzed is the entry-level bin of the Atlas 200I A2 family — a 0.5 GHz AI core, 1.0 GHz CPU cluster, delivering 8 TOPS INT8. Most published benchmarks, specs, and comparisons for "Atlas 200I A2" online describe the 20 TOPS variant, not this one, so any performance comparison should explicitly confirm which Board ID the source material refers to before drawing conclusions.